The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice*

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(Dated: March 10, 2026)

Fock showed in 1935 that stereographic projection turns the momentum-space hydrogen Schrödinger equation into a rotation-invariant integral equation on the unit three-sphere S^3, whose kernel $1/|\omega-\omega'|^2$ is zonal and diagonal on the hyperspherical harmonics (the eigenfunctions of the Laplace–Beltrami operator Δ_{S^3}, eigenvalue $-(n^2-1)$). **[PRIOR ART]** On those harmonics the kernel acts as $2\pi^2(1-\Delta_{S^3})^{-1/2}$ (eigenvalue $2\pi^2/n$), so the bound-state Hamiltonian is a function of Δ_{S^3}, $H=-\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$, i.e. $E_n=-Z^2/(2(C+1))$ with $C=n^2-1$ the $SO(4)$ Casimir: in this sense Fock's projection makes Δ_{S^3} equivalent to the Schrödinger problem. (The kernel is the restriction to S^3 of the Green's function of the $\mathbb{R}^4$ Laplacian, not the Green's function of Δ_{S^3}.) **[MEASURED]** We report what the GeoVac discrete graph Laplacian is measured to do through $n_{\max}=30$— **[INTERNAL THEOREM]** its top eigenvalue saturates the bipartite bound $\lambda_{\max}\to 2d_{\max}=8$ at the closed-form rate $(C_{\rm sat}+o(1))/n_{\max}^2$ (Paper 0 §VI; $C_{\rm sat}$ a saturation constant, not the Casimir C), proved rather than sampled—but this saturation is the band edge of two-dimensional grid graphs (twice the maximum degree) and carries no S^3 information. **[WALL]** The edge Laplacian $L=D-A$ does *not* converge to the S^3 Laplace–Beltrami operator: its edges never change l, so it splits into $n_{\max}$ disconnected l-blocks, and its kernel is $n_{\max}$-dimensional at every cutoff while Δ_{S^3} has a single zero mode (`tests/test_graph_s3_nonconvergence.py`). What survives is exact at every cutoff: the node set (n,l,m), n^2 per shell, is the Gel'fand–Tsetlin label set of the Peter–Weyl truncation of $L^2(S^3)$, on which $-\Delta_{S^3}=\mathbf{L}^2+\mathbf{A}^2$ acts as the diagonal $SO(4)$ Casimir n^2-1 (a node weight, not an edge Laplacian; standard $SO(4)$ representation theory). The S^3 equivalence itself is carried by Fock's continuum projection, whose chain we audit by symbolic computation (its non-definitional step, the kernel eigenvalue, is tested for $n\le 4$; general n is Fock's). **[PRIOR ART]** The Laplace–Beltrami operator on the round unit S^3 has pure integer eigenvalues $\lambda_n=-(n^2-1)$ with no intrinsic physical scale (standard; checked symbolically for $n\le 4$). **[SYMBOLIC PROOF]** The Rydberg energy levels $E_n=-Z^2/(2n^2)$ enter only through the energy-shell constraint $p_0^2=-2E$, which acts as a "stereographic focal length" mapping the unit sphere to flat $\mathbb{R}^3$ momentum coordinates; the quantization $p_0=Z/n$ itself is forced by the eigenvalue $2\pi^2/n=\mathrm{Vol}(S^3)/n$ of Fock's S^3 Coulomb kernel on the degree-$(n-1)$ harmonics. **[PRIOR ART]** That eigenvalue holds for every n (Fock 1935; the Funk–Hecke theorem); we verify it in exact arithmetic for $n\le 4$ and every $l<n$ (one real harmonic per l) without inputting E_n (`tests/test_paper7_fock_kernel.py`). **[SYMBOLIC PROOF + MEASURED]** At positive energy the projection lands on the hyperboloid H^3: continuing $n\to iZ/k$ releases only the cap $l\le n-1$, and every continuum density tested is the continued bound one times the single factor $1/(1-e^{-2\pi Z/k})$, the ratio of the two normalizations (Section V F, `tests/test_paper7_hydrogen_continuum_door.py`); the factor is standard quantum defect theory [11]. At $E=0$, the third shape, the Runge–Lenz algebra contracts to the Euclidean group and both sides meet one π-free threshold function. **[SYMBOLIC PROOF]** The projection geometry is audited by a suite of 18 symbolic checks (ten of them limiting-case, definitional or non-discriminating) covering the stereographic map, the conformal Laplacian and its $l=0$ spectrum, and the (definitional) eigenvalue-to-energy bookkeeping.

I. INTRODUCTION

The Schrödinger equation is the foundation of non-relativistic quantum mechanics. In atomic units ($\hbar=m_e=e=1$), the time-independent equation for hydrogen reads

$$\left(-\frac{1}{2}\nabla^2-\frac{1}{r}\right)\psi(\mathbf{r})=E\,\psi(\mathbf{r}). \tag{1}$$

This equation presupposes a continuous, flat Euclidean space $\mathbb{R}^3$ and a radial Coulomb potential $V(r)=-1/r$ with explicit physical dimensions. These assumptions are so deeply embedded in the formalism that they are rarely questioned: continuous space provides the arena, the Coulomb potential provides the dynamics, and the quantized energy levels $E_n=-1/(2n^2)$ emerge as boundary-condition eigenvalues.

Yet a remarkable observation, first made by Fock in 1935 [1], suggests a fundamentally different perspective. Fock showed that when hydrogen is analyzed in momentum space, the Schrödinger equation acquires an unexpected geometric structure. The momentum-space integral equation, with its $1/|\mathbf{p}-\mathbf{p}'|^2$ kernel (the Fourier transform of the Coulomb potential), can be mapped by stereographic projection onto the three-sphere S^3. On

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S^3, the Coulomb problem becomes a rotation-invariant integral equation whose kernel $1/|\boldsymbol{\omega}-\boldsymbol{\omega}'|^2$ is zonal, with eigenvalue $2\pi^2/n$ on the degree-$(n-1)$ hyperspherical harmonics (the eigenfunctions of Δ_{S^3}); that eigenvalue forces the focal length $p_0 = Z/n$. Since the degree-$(n-1)$ harmonics are the eigenfunctions of Δ_{S^3} with eigenvalue $-(n^2-1)$, the kernel acts on them as $2\pi^2(1-\Delta_{S^3})^{-1/2}$, and the bound-state Hamiltonian is itself a function of Δ_{S^3}, $H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$ ($E_n = -Z^2/(2(C+1))$, $C = n^2-1$): this is the precise sense of the standard "free motion on S^3" reading. The kernel is not the Green's function of Δ_{S^3} (that behaves like $1/d$, with eigenvalues $\propto 1/(n^2-1)$); it is the restriction to S^3 of the $\mathbb{R}^4$ Laplacian's Green's function (Section VIII). In Fock's formulation the $1/r$ potential appears as a projection effect of that curved geometry viewed in flat coordinates—an equivalent description, not a claim that either one is more fundamental.

This observation was developed further by Bargmann [4], who connected the hidden $SO(4)$ symmetry of hydrogen to the isometry group of S^3, and by Malkin and Man'ko [7], Fronsdal [8] and Barut and Kleinert [5], who extended the symmetry to the conformal group $SO(4,2)$. The mathematical structure has been thoroughly studied in the context of dynamical symmetry groups [12, 14], and the Coulomb-Sturmian and hyperspherical-harmonic literature has long exploited it computationally (e.g. Avery [3]).

In this paper we emphasize that structural content. The S^3 representation carries exact discrete structure — the labels, degeneracies and selection rules of the S^3 harmonics — that can be exploited computationally, independently of any reading of which description is physically prior. Our starting point is the GeoVac framework [15, 16], which indexes the hydrogen state space by a discrete label graph: nodes label the quantum states $|n,l,m\rangle$, edges connect labels related by the $L_\pm$ and $T_\pm$ ladder operators, and the Hamiltonian matrices built on that label set are sparse (Section III states exactly which of the graph's properties carry S^3 content and which do not). This discrete lattice is *intrinsically dimensionless*—it is a pure combinatorial topology with no inherent length, time, or energy scales.

Its central results are one spectral-bound statement (proved in closed form, Paper 0 §VI) and one wall (the edge Laplacian does not converge to Δ_{S^3}); items 2 and 3 are classical facts it builds on, re-verified symbolically:

1. **[INTERNAL THEOREM]** The top Laplacian eigenvalue of the discrete graph saturates its bipartite bound, $\lambda_{\max} \to 2d_{\max} = 8$, at the closed-form rate $(C_{\text{sat}} + o(1))/n_{\max}^2$ (Paper 0 §VI; measured through $n_{\max} = 30$ and 70). **[WALL]** This is the band edge of two-dimensional grid graphs, not an indication of S^3: L splits into $n_{\max}$ disconnected l-blocks and does not converge to Δ_{S^3} (see New Contributions item 1). What is exact is the label set: the nodes (n,l,m) index the Peter–Weyl truncation of $L^2(S^3)$, on which $-\Delta_{S^3} = \mathbf{L}^2 + \mathbf{A}^2$ acts as the diagonal $SO(4)$ Casimir.
2. **[PRIOR ART; SYMBOLIC PROOF]** The Laplace-Beltrami operator on this unit sphere produces pure integer eigenvalues $\lambda_n = -(n^2-1)$, carrying no physical dimensions.
3. **[SYMBOLIC PROOF]** The dimensionful Rydberg spectrum $E_n = -Z^2/(2n^2)$ enters only through the energy-shell constraint $p_0^2 = -2E$, which serves as a projection parameter mapping S^3 coordinates to flat $\mathbb{R}^3$ coordinates; the value $p_0 = Z/n$ is forced by the eigenvalue $2\pi^2/n$ of Fock's S^3 Coulomb kernel (Section V D; **[PRIOR ART]** the eigenvalue holds for every n, Fock 1935 / Funk–Hecke; verified in exact arithmetic for $n \le 4$, all $l < n$, one real harmonic per l, with E_n never an input, `tests/test_paper7_fock_kernel.py`). The rearrangement $E_n = -p_0^2/2$ by itself is definitional.

These claims are supported by a suite of 18 symbolic checks, implemented in `sympy` and verified computationally, that audit the algebraic steps of the conformal transformation chain (several are limiting-case or definitional substitutions rather than logically independent theorems), together with the kernel-eigenvalue and eigenfunction checks of `tests/test_paper7_fock_kernel.py`, which carry the non-definitional part of the energy step.

The structure of this paper is as follows. Section III describes the dimensionless graph topology and its relationship to S^3. Section IV details the stereographic projection and the role of the conformal factor. Section V states the operator equivalence: through Fock's integral equation the bound-state Hamiltonian is the function $H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$ of the S^3 Laplacian, its scale $p_0 = Z/n$ fixed by the kernel eigenvalue (Section V D); the local conformal-Laplacian identity recorded there is bookkeeping, and its appendix check is non-discriminating. Section VI rewrites the two-electron repulsion integral in its standard Fourier–Parseval form, in the stereographic variable $t = q/(2Z)$, as a density-overlap integral **[PRIOR ART]**. Section VIII discusses the physical and philosophical implications.

II. WHAT IS NEW VS. WHAT IS KNOWN

The S^3 structure of the hydrogen atom has a distinguished history, and we emphasize what is established mathematics versus what is new in this work.

A. Known Results (Prior Work)

The following results are well-established in the mathematical physics literature:

- *Fock (1935)* [1]: The hydrogen atom in momentum space maps to S^3 via stereographic projection. The

Coulomb problem becomes a rotation-invariant integral equation on the sphere, with the $1/r$ potential emerging from the chordal distance identity; its $SO(4)$ symmetry accounts for the n^2 degeneracy (the standard "free particle on S^3" reading is exact in the sense that the bound states are the Δ_{S^3} eigenfunctions and the Hamiltonian is the function $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$ of that operator, the kernel eigenvalue $2\pi^2/n$ fixing $p_0 = Z/n$).

- *Bargmann (1936)* [4]: The hidden $SO(4)$ symmetry of hydrogen corresponds to the isometry group of S^3. The integer spectrum $\lambda_n = -(n^2 - 1)$ follows from the representation theory of $SO(4)$.

- *Malkin and Man'ko (1965), Fronsdal (1967), Barut and Kleinert (1967)* [5–8]: The full conformal group $SO(4, 2)$ acts as the dynamical symmetry group of hydrogen, connecting S^3 to the non-compact spectrum-generating algebra.

The stereographic projection formulas (Eqs. 7–8), the unit sphere constraint (Eq. 9), the chordal distance identity (Eq. 14), the conformal Laplacian relation (Eq. 17), and the integer eigenvalue spectrum (Eq. 26) are all known results. Our symbolic checks in the Appendix verify these identities; they do not claim to derive them for the first time.

B. New Contributions of This Work

The following are the specific new contributions:

1. **Discrete graph: label lattice, not a discretization (non-convergence). [WALL]** The GeoVac discrete paraboloid lattice—with nodes (n, l, m) and edges generated by $L_\pm, T_\pm$ operators—is *not* a discretization whose continuum limit is the Laplace-Beltrami operator on unit S^3. Its edges never change l, so $L = D - A$ (binary production default and the $1/(n_i n_j)$ option alike) is the direct sum of $n_{\max}$ disconnected grid graphs $P_{n_{\max}-l} \times P_{2l+1}$ with free ends: $\dim\ker L = n_{\max}$ where Δ_{S^3} has a single zero mode (`tests/test_graph_s3_nonconvergence.py`); this kernel count is the load-bearing obstruction. (The graph heat kernel e^{-tL} also vanishes identically between blocks; reading that as a statement about the continuum heat kernel would need a node-to-point identification this paper does not make, so it corroborates rather than carries the argument.) This wall is the new content of the item. **[PRIOR ART]** What *is* exact, at every cutoff, is the label set: the nodes (n, l, m), n^2 per shell, are the Gel'fand–Tsetlin labels of the Peter–Weyl truncation of $L^2(S^3)$, and on them $-\Delta_{S^3}$ acts as the diagonal $SO(4)$ Casimir $\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1$ (**A** the energy-normalized Runge–Lenz vector), with multiplicity n^2 on every shell including the top one (standard $SO(4)$ representation theory [2, 4], the shells being complete irreducible representations; built from the generators and checked in the same test). On the label lattice Δ_{S^3} is a node weight, not an edge Laplacian. What is **[MEASURED]** of L itself through $n_{\max} = 30$ (`tests/test_paper7_graph_convergence.py`; `tests/test_trunk_qa_splitting.py`, Paper 1) is: (i) **[INTERNAL THEOREM]** the top Laplacian eigenvalue of the production lattice saturates the bipartite bound, $\lambda_{\max} \to 2d_{\max} = 8$ — the ℓ-block grid structure makes the spectrum closed form, so this is proved rather than sampled — (6.618 at $n_{\max} = 5$, 7.954 at 30), at the closed-form rate $(C_{\text{sat}} + o(1))/n_{\max}^2$ with $C_{\text{sat}} = \frac{\pi^2}{4}(2 + 2^{1/3})^2(1 + 2^{-2/3}) = 42.7397\ldots$ that the ℓ-block grid structure supplies ([**MEASURED**] every sample checked understates C_{sat} — all 591 integer cutoffs $10 \le n_{\max} \le 600$, Paper 0 §VI, `tests/test_lattice_spectrum.py` — a one-sided pattern, measured, not proved, and *not* monotonicity: the approach is not monotone) (Paper 0 §VI: each ℓ-block is the grid graph $P_{n_{\max}-\ell} \times P_{2\ell+1}$, so the spectrum is closed form), so that with the matched scale $\kappa = -1/16$ the extremal eigenvalue $\kappa\lambda_{\max}$ approaches $-1/2$ *by construction*— the content is the saturation, not the target; (ii) the $2s/2p$ node-amplitude proxy on the binary lattice decays, oscillating, to 0.39% at $n_{\max} = 30$ (on the favourable $n_{\max} \equiv 0 \pmod 3$ branch, where $\lambda_{2s} = 3$ exactly; the neighbours 28, 29 are 4–7× larger; a proxy, not a spectral gap: the ℓ-blocks are disconnected and the mode selection is a near-tie — Paper 1 §III conditioning caveat). The $\lambda_{\max}$ mode lives in a mid-l block with negligible weight on the $1s$ node (the $l = 0$ block alone saturates only $\lambda_{\max} \to 4$), and $8 = 2d_{\max}$ is the band edge of any large two-dimensional grid, carrying no S^3 information (the constant C_{sat} is a Dirichlet band edge); the spectrum of κL is confined to $[-1/2, 0]$, so no level-by-level Rydberg ladder is reproduced. The operator convergence $L \to \Delta_{S^3}$ is therefore not merely unestablished but false.

2. **Explicit symbolic verification. [SYMBOLIC PROOF]** The 18 symbolic checks (Appendix A) provide a machine-verified algebraic audit of the conformal transformation chain (ten of the eighteen checks are definitional, limiting-case or non-discriminating rather than logically independent theorems). While the underlying identities are known, this systematic verification in `sympy` is new and provides a reproducible computational check.

3. **Scale-invariance framing (organizing use only).** We articulate the "dimensionless vacuum" perspective: the observation that the con-

formal factor Ω always produces the *unit* S^3 regardless of p_0, and that all physical dimensions enter exclusively through the energy-shell constraint. **[PRIOR ART]** The property itself is Fock's, and it is explicit in the group-theoretic literature: Bander and Itzykson rescale momentum by $\sqrt{2\mu|E|}$ and project onto the *unit* sphere (bound states) or unit hyperboloid (scattering) [12, 13]. What this paper adds is only its use as an organizing principle for the GeoVac label-set approach.

4. **Computational methodology. [MEASURED]** The practical consequence: replacing continuous integration with sparse Hamiltonian-matrix eigenvalue problems whose operator size is near-linear in V (adjacency nonzeros scale as $V^{1.07}$; the timing is guarded sub-quadratic, Paper 1). What this paper supplies to that approach is the exact label set (n, l, m) and the selection rules it carries, together with Fock's continuum equivalence; it does *not* supply an equivalence between the edge Laplacian and Δ_{S^3} (item 1).

5. **The continuum as a released label. [SYMBOLIC PROOF + MEASURED]** Continuing $n \to iZ/k$ carries Fock's sphere onto Bander and Itzykson's hyperboloid, keeps l, m and the Gaunt rules, and releases only the cap $l \leq n-1$; every continuum density tested is the continued bound one times the single factor $1/(1-e^{-2\pi Z/k})$, the ratio of the two normalizations (Section V F); at $E = 0$ the symmetry contracts to the Euclidean group and both sides meet one π-free threshold function. **[PRIOR ART]** Stobbe's formula and the Sommerfeld factor are classical, and the single factor and the threshold limit are quantum defect theory [11]; what is added is the accounting of what the release keeps and what it lets in.

III. THE DIMENSIONLESS GRAPH

A. The Paraboloid Lattice

The GeoVac framework indexes the single-electron hydrogen state space by a finite graph $\mathcal{G} = (V, E)$. The vertex set V consists of the labels of all quantum states $|n, l, m\rangle$ with $1 \leq n \leq n_{\max}$, $0 \leq l < n$, and $-l \leq m \leq l$, giving $|V| = \sum_{n=1}^{n_{\max}} n^2$ nodes. The edge set E is generated by four transition operators:

$$L_\pm |n, l, m\rangle \propto |n, l, m \pm 1\rangle, \tag{2}$$

$$T_\pm |n, l, m\rangle \propto |n \pm 1, l, m\rangle. \tag{3}$$

These operators form the algebraic skeleton of the $SU(2) \otimes SU(1,1)$ dynamical symmetry group, a compact $\times$ non-compact subgroup of the hydrogen conformal group $SO(4,2)$ (whose maximal compact subgroup is $SO(4) \times SO(2)$) [5].

The graph Laplacian is defined as

$$L = D - A, \tag{4}$$

where A_{ij} is the weighted adjacency matrix constructed from the transition operators and D is the diagonal degree matrix with entries $D_{ii} = \sum_j |A_{ij}|$. The (legacy) graph Hamiltonian is

$$H = \kappa\, L = \kappa(D - A), \tag{5}$$

where $\kappa = -1/16$ is the kinetic scale factor. It is not the physical hydrogen Hamiltonian, which on these labels is diagonal, $-1/(2n^2)$. **[OBSERVATION]** κ matches the graph's extremal eigenvalue (a band edge, $\lambda_{\max} \to 8$ on the binary graph) to the ground-state energy $-\frac{1}{2}$; that $8 = 2d_{\max}$ is the band edge of two-dimensional grids, not an S^3 quantity, and the rest of the spectrum of κL is not the Rydberg ladder (it lies in $[-\frac{1}{2}, 0]$). The per-shell energies E_n come instead from Fock's continuum projection: the eigenvalue $2\pi^2/n$ of his S^3 Coulomb kernel forces the focal length $p_0 = Z/n$, and the energy shell $p_0^2 = -2E$ then gives $E_n = -Z^2/(2n^2)$; the label Casimir $C = n^2 - 1$ alone does not fix the function, but once Fock's kernel fixes it, $E_n = -Z^2/(2(C+1))$ on the labels, i.e. $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$ (Section V D). **[OBSERVATION]** This matching value *coincides* with a single conformal-geometry value, $1/16$, namely the fourth power $1/\Omega^4(0)$ of the inverse Fock conformal factor, with $\Omega(0) = 2$ (Section VIII; equivalently one quarter of the s-wave coupling $c^2(n, 0) = 1/4$, the Chebyshev amplitude being $1/2$) in the dimensionless $p_0 = 1$ convention used throughout this section (dimensionfully $\Omega(0) = 2/p_0$). We record this as an *observation*: the matching κ and the geometric $1/16$ agree numerically, but are computed from disjoint inputs and no mechanism is known that derives one from the other.

B. Intrinsic Dimensionlessness

A crucial observation about the graph $\mathcal{G}$ is that it carries *no intrinsic physical scale*. The quantum numbers (n, l, m) are pure integers. The edge weights are dimensionless: binary ($w_{ij} = 1$) in the production default, or $w_{ij} \propto 1/(n_i n_j)$ under the `topological_weights` option. The graph Laplacian L is a dimensionless matrix. The only dimensionful quantity in Eq. (5) is the kinetic scale factor κ, which assigns an energy unit to the graph's extremal eigenvalue. But κ is an *external* parameter imposed at the measurement stage—it is not encoded in the graph topology itself.

This dimensionlessness is not accidental. Its S^3 content lies in the *node set*, not in the edge Laplacian: the labels (n, l, m) are the finite Gel'fand–Tsetlin index set of the unit three-sphere's harmonics. To see what does and does not carry over, we examine the large-$n_{\max}$ behaviour.

C. The Label Lattice and the Unit S^3 (No Continuum Limit of L)

The node set of $\mathcal{G}$ is exactly the label set (n, l, m), n^2 per shell, of the Peter–Weyl truncation of $L^2(S^3)$ at $n \le n_{\max}$, at every cutoff. On these labels the Laplace–Beltrami operator acts exactly: $-\Delta_{S^3} = \mathbf{L}^2 + \mathbf{A}^2$ is the diagonal $SO(4)$ Casimir $n^2 - 1$ (a node weight). **[WALL]** The edge Laplacian L does *not* converge to Δ_{S^3} as $n_{\max} \to \infty$: it preserves l, splits into $n_{\max}$ disconnected blocks (kernel dimension $n_{\max}$ against Δ_{S^3}'s one zero mode), and its one saturating spectral feature, $\lambda_{\max} \to 8$, is the band edge of two-dimensional grids (New Contributions, item 1; the s/p proxy below, whose size tracks the residue of $n_{\max}$ modulo 3, is not evidence either).

At finite $n_{\max}$, the graph Laplacian exhibits discretization artifacts: the $2s/2p$ node-amplitude proxy reaches 37% at $n_{\max} = 5$ on the binary production lattice (the splitting is strongly non-monotone: 12.7%, 0.65%, 15.6%, 5.1% and 1.7% at $n_{\max} = 6, 7, 8, 9, 10$, then 4.1% at 20 and 0.39% at 30), **[INTERNAL THEOREM]** set by the residue of $n_{\max}$ modulo 3 [16]; the sectors' degrees do differ ($D_{2p,\, m=0} = 3 > D_{2s} = 2$; the $m = \pm 1$ nodes tie), but the l-blocks are disconnected components, so no operator acts across them to lift a degeneracy, and a degree difference is not a mechanism for the proxy. However, this splitting oscillates and decays systematically:

$$\Delta E_{2s\text{-}2p}(n_{\max}) \to 0 \quad \text{as} \quad n_{\max} \to \infty, \tag{6}$$

the two proxy eigenvalues coinciding in the limit — a path-graph coincidence, *not* a reproduction of the continuum $SO(4)$ Coulomb degeneracy (the ℓ-blocks are disconnected). **[INTERNAL THEOREM]** The rate is closed form and is governed by the residue of $n_{\max}$ modulo 3, not by proximity to a continuum limit: $\lambda_{2s} = 3$ exactly when $n_{\max} \equiv 0 \pmod 3$, where the ratio is $\big(2 - 2\cos\frac{\pi}{n_{\max}-1}\big)/3 = O(n_{\max}^{-2})$, while off that class it is $O(n_{\max}^{-1})$, so at neighbouring cutoffs the off-branch value is generically larger — asymptotically by $n_{\max}/(\sqrt{3}\,\pi)$, but far from that at small cutoff: 4.2× and 6.6× at $n_{\max} = 28, 29$ against 30 (1.65%, 2.58% vs 0.39%), where the formula gives 5.5; 20–23× at 118, 119 against 120 (22.1); 550–552× at 2998, 2999 against 3000 (551.3). Up to $n_{\max} = 13$ the ordering can invert (0.65% at 7 against 12.7% at 6; 2.17% at 13 against 2.70% at 12) (Paper 1 §III for the full table; the factors at 118–120 and 2998–3000 are pinned in closed form by `tests/test_paper1_block_spectrum.py`). The ℓ-blocks are disconnected, so no degeneracy is present for the proxy to lift (Paper 1 §III). **This leg accordingly carries no evidential weight for a continuum reading**, and neither does the $\lambda_{\max}$ saturation (a grid band edge). **[MEASURED]** At $n_{\max} = 30$ the proxy is 0.39%, on the $n_{\max} \equiv 0 \pmod 3$ branch; at 28 and 29 it is 1.65% and 2.58%.

The exact degeneracy *of the continuum operator* is a hallmark of $SO(4)$ symmetry—the isometry group of S^3. **[MEASURED]** The graph's finite-$n_{\max}$ proxy for it is not: its value tracks the residue of $n_{\max}$ modulo 3 (Paper 1 §III), so its decay is not a recovery of that symmetry. **[WALL]** The production graph is disconnected into $n_{\max}$ components (one per l), so no continuum limit of L carries the $SO(4)$ isometry; the $SO(4)$ structure lives instead in the label set, where the Casimir $n^2 - 1$ is exact and n^2-fold on every shell (`tests/test_graph_s3_nonconvergence.py`).

The critical question is: what is the *radius* of this sphere? As we shall demonstrate in Section IV, the answer is unambiguous: the intrinsic geometry is always that of the *unit* S^3, regardless of any physical parameters. This is the mathematical content of scale invariance.

IV. THE CONFORMAL LENS: STEREOGRAPHIC PROJECTION

A. Fock's Mapping

Fock's 1935 insight was to map three-dimensional flat momentum space $\mathbb{R}^3$ onto the unit three-sphere $S^3 \subset \mathbb{R}^4$ via stereographic projection [1]. Given a momentum vector $\mathbf{p} = (p_1, p_2, p_3)$ and an energy-shell parameter p_0 (defined below), the projection is:

$$n_i = \frac{2p_0\, p_i}{p^2 + p_0^2}, \quad i = 1, 2, 3, \tag{7}$$

$$n_4 = \frac{p^2 - p_0^2}{p^2 + p_0^2}, \tag{8}$$

where $p^2 = p_1^2 + p_2^2 + p_3^2$. This map sends the origin $\mathbf{p} = 0$ to the south pole $(0, 0, 0, -1)$ and the point at infinity $|\mathbf{p}| \to \infty$ to the north pole $(0, 0, 0, +1)$, thereby compactifying $\mathbb{R}^3$ into S^3.

B. The Unit Sphere Constraint

A direct algebraic computation confirms that the image lies on the unit sphere:

$$\sum_{i=1}^{4} n_i^2 = \frac{4p_0^2 p^2 + (p^2 - p_0^2)^2}{(p^2 + p_0^2)^2} = 1. \tag{9}$$

This identity holds for *all* values of $p_0 > 0$. The parameter p_0 does not change the target manifold; it only changes which point in $\mathbb{R}^3$ maps to the south pole. **[SYMBOLIC PROOF]** This classical property of stereographic projection, re-verified here, is the starting point:

The stereographic projection always produces the unit S^3, regardless of p_0. The intrinsic geometry of the target manifold has no physical scale.

To make this explicit, define the conformal factor

$$\Omega(\mathbf{p}) = \frac{2p_0}{p^2 + p_0^2}. \tag{10}$$

The spatial components of the projection satisfy the identity

$$n_i = \Omega \cdot p_i, \quad i = 1, 2, 3, \tag{11}$$

confirming that Ω is the local scale factor of the map. Under the substitution $u = p/p_0$ (dimensionless momentum), the conformal factor becomes

$$\Omega = \frac{2}{p_0(1+u^2)}, \tag{12}$$

and the induced metric on S^3, expressed in flat stereographic coordinates, takes the form

$$\mathrm{d}s^2_{S^3} = \Omega^2 \, \mathrm{d}p^2 = \frac{4}{(1+u^2)^2} \, \mathrm{d}u^2. \tag{13}$$

This is the standard round metric on the unit S^3 in stereographic coordinates. **[SYMBOLIC PROOF]** The factors of p_0 cancel identically, confirming that the intrinsic geometry is scale-invariant.

C. The Chordal Distance Identity

The conformal structure of the projection is captured by a fundamental distance identity. For two momentum vectors $\mathbf{p}$ and $\mathbf{q}$ with images $\mathbf{n}(\mathbf{p})$ and $\mathbf{n}(\mathbf{q})$ on S^3:

$$|\mathbf{n}(\mathbf{p}) - \mathbf{n}(\mathbf{q})|^2 = \Omega(\mathbf{p}) \, \Omega(\mathbf{q}) \, |\mathbf{p} - \mathbf{q}|^2. \tag{14}$$

The left side is the chordal (Euclidean 4D) distance on S^3; the right side is the flat Euclidean distance in $\mathbb{R}^3$, weighted by conformal factors at both endpoints.

This identity is the algebraic engine behind the emergence of the Coulomb potential. The momentum-space Schrödinger equation contains the kernel $1/|\mathbf{p} - \mathbf{p}'|^2$, which is the Fourier transform of the $1/r$ Coulomb potential. Under Fock's projection, Eq. (14) converts this kernel into $1/|\mathbf{n} - \mathbf{n}'|^2$ on S^3 (up to conformal weights), transforming the Coulomb problem into a rotation-invariant integral equation on the sphere [1], Eq. (28), whose kernel eigenvalue $2\pi^2/n$ forces $p_0 = Z/n$ (Section V D). On the harmonics the kernel acts as $2\pi^2(1 - \Delta_{S^3})^{-1/2}$, so the problem is a function of Δ_{S^3} (the Hamiltonian is $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$); the kernel is not, however, the Green's function of Δ_{S^3} (Section VIII).

D. Physical Interpretation of p_0

What is p_0? In Fock's original treatment, it is defined by the energy-shell constraint:

$$p_0^2 = -2mE, \tag{15}$$

where $E < 0$ for bound states (we set $m = 1$ in atomic units). Its allowed values are not an input: Fock's S^3 Coulomb kernel forces $p_0 = Z/n$ (Section V D), i.e. $p_0 = 1/n$ for hydrogen.

Crucially, p_0 does *not* determine the curvature of the target manifold—the unit sphere constraint Eq. (9) holds for all p_0. Instead, p_0 acts as a **stereographic focal length**: it determines how the flat momentum coordinates $\mathbf{p}$ are laid out on the fixed unit sphere. Different values of p_0 correspond to different stereographic charts of the *same* underlying manifold, analogous to choosing different focal lengths for a camera lens viewing the same scene.

The conformal factor Ω encodes this projection:

- At $\mathbf{p} = 0$: $\Omega = 2/p_0$ (maximum magnification at the south pole).
- As $|\mathbf{p}| \to \infty$: $\Omega \to 0$ (the north pole is infinitely compressed).

The volume element transforms as $\mathrm{d}\Omega_{S^3} = \Omega^3 \, \mathrm{d}^3p$, where the Jacobian Ω^3 encodes the non-uniform compression of flat space onto the sphere.

E. Inverse Projection

The map is invertible away from the north pole:

$$p_i = \frac{p_0 \, n_i}{1 - n_4}, \quad i = 1, 2, 3. \tag{16}$$

This confirms the projection is a diffeomorphism $S^3 \setminus \{N\} \to \mathbb{R}^3$, where $N = (0, 0, 0, 1)$ is the north pole. The compactification of $\mathbb{R}^3$ to S^3 adjoins this single point at infinity, mirroring the Riemann sphere compactification $\mathbb{C} \cup \{\infty\} \cong S^2$ in one lower dimension.

V. OPERATOR EQUIVALENCE ON THE CONTINUUM S^3

A. The Conformal Laplacian Identity

Having established the geometric setting, we now demonstrate how the Laplace-Beltrami operator on S^3—the continuum operator whose harmonics the graph's nodes label (it is *not* a limit of the graph's edge Laplacian, Section III)—relates to the flat-space Schrödinger operator. The equivalence proper is Fock's, Section V D: the bound-state Hamiltonian transported to S^3 is the function $H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$. The conformal-Laplacian identity below is the local bookkeeping of the projection, and the appendix check made of it (test 15) is non-discriminating.

For a conformal rescaling of the metric $g_{S^3} = \Omega^2 g_{\mathrm{flat}}$ in $n = 3$ dimensions, the Laplace-Beltrami operators are related by

$$\Delta_{S^3} f = \Omega^{-2} \left[\Delta_{\mathrm{flat}} f + \nabla(\ln \Omega) \cdot \nabla f \right], \tag{17}$$

where $\Delta_{\text{flat}} = \partial^2/\partial p^2 + (2/p)\,\partial/\partial p$ is the radial Laplacian in three-dimensional flat space (for the $l = 0$ sector), and the connection correction $\nabla(\ln\Omega)\cdot\nabla f$ encodes the curvature of the stereographic coordinate system.

For the conformal factor $\Omega = 2p_0/(p^2+p_0^2)$, the logarithmic gradient is

$$\frac{\mathrm{d}\ln\Omega}{\mathrm{d}p} = -\frac{2p}{p^2+p_0^2}. \tag{18}$$

This is the *connection term* that distinguishes the curved Laplacian from the flat one. It is a rational function of p—no transcendental functions appear—which is essential for clean algebraic manipulation.

B. Verification: Constants and Eigenfunctions

As a consistency check, the Laplace-Beltrami operator annihilates constants:

$$\Delta_{S^3}(1) = 0, \tag{19}$$

confirming the basic property that constants are harmonic on any Riemannian manifold.

For the first nontrivial eigenfunction, consider the hyperspherical angle χ defined by the stereographic projection $p = p_0\tan(\chi/2)$. The function $\cos\chi$, which in stereographic coordinates takes the form

$$\cos\chi = \frac{p_0^2 - p^2}{p_0^2 + p^2}, \tag{20}$$

is the $n = 2$, $l = 0$ eigenfunction of Δ_{S^3}. Direct symbolic computation confirms

$$\Delta_{S^3}(\cos\chi) = -3\,\cos\chi. \tag{21}$$

More generally, the $l = 0$ eigenfunctions are Gegenbauer polynomials $C^1_{n-1}(\cos\chi)$:

$$C^1_0 = 1, \tag{22}$$
$$C^1_1 = 2\cos\chi, \tag{23}$$
$$C^1_2 = 4\cos^2\chi - 1, \tag{24}$$

with eigenvalues

$$\Delta_{S^3}\, C^1_{n-1}(\cos\chi) = -(n^2-1)\, C^1_{n-1}(\cos\chi). \tag{25}$$

[SYMBOLIC PROOF] We have verified Eq. (25) symbolically for $n = 1, 2, 3$ using the conformal Laplacian formula, confirming exact agreement. Independently, `tests/test_paper7_fock_kernel.py` checks that every Fock harmonic $\sin^l\chi\, C^{l+1}_{n-l-1}(\cos\chi)\, Y_l$ with $n \le 4$ and $l < n$ (one real harmonic per l), built as a polynomial on $\mathbb{R}^4$, is harmonic and homogeneous of degree $n-1$, hence an eigenfunction of Δ_{S^3} with eigenvalue $-(n^2-1)$, including $l > 0$.

C. The Dimensionless Spectrum

Equation (25) is the central result of this section. The eigenvalues of the Laplace-Beltrami operator on the unit S^3 are

$$\lambda_n = -(n^2-1), \quad n = 1, 2, 3, \ldots \tag{26}$$

These are *pure integers*. They carry no physical dimensions. They depend on no physical parameters—not on p_0, not on the electron mass, not on the fine structure constant. They are properties of the round unit-sphere geometry alone (the metric, not merely the topology, fixes them).

This spectrum arises from the same mechanism as the quantization of angular momentum on S^2: the compactness of the manifold forces the eigenfunctions to satisfy periodicity constraints that admit only discrete mode numbers. On S^3, these mode numbers correspond to the principal quantum number n of hydrogen.

D. The Energy Shell: Projecting to Physics

How, then, do the dimensionful hydrogen energy levels emerge from dimensionless eigenvalues? Two ingredients are needed, and only one of them is definitional.

The quantization (not definitional). In momentum space the Schrödinger equation for charge Z reads

$$\left(\frac{p^2}{2} - E\right)\varphi(\mathbf{p}) = \frac{Z}{2\pi^2}\int \frac{\varphi(\mathbf{p}')}{|\mathbf{p}-\mathbf{p}'|^2}\,\mathrm{d}^3p', \tag{27}$$

the prefactor $Z/(2\pi^2)$ being the Fourier transform of $-Z/r$. With $p_0^2 = -2E$ and $\varphi = \Omega^2\Psi$, the chordal identity Eq. (14) and the Jacobian $\mathrm{d}\Omega_{S^3} = \Omega^3\,\mathrm{d}^3p$ reduce both sides to a common factor Ω times

$$p_0\,\Psi(\boldsymbol{\omega}) = \frac{Z}{2\pi^2}\int_{S^3}\frac{\Psi(\boldsymbol{\omega}')}{|\boldsymbol{\omega}-\boldsymbol{\omega}'|^2}\,\mathrm{d}\Omega'. \tag{28}$$

The kernel $1/|\boldsymbol{\omega}-\boldsymbol{\omega}'|^2$ depends only on $\boldsymbol{\omega}\cdot\boldsymbol{\omega}'$, so it is diagonal on the S^3 harmonics; on the degree-$(n-1)$ harmonics its eigenvalue is

$$\int_{S^3}\frac{Y_{nlm}(\boldsymbol{\omega}')}{|\boldsymbol{\omega}-\boldsymbol{\omega}'|^2}\,\mathrm{d}\Omega' = \frac{2\pi^2}{n}\,Y_{nlm}(\boldsymbol{\omega}), \tag{29}$$

independent of l and m. **[PRIOR ART]** This holds for every n: it is Fock's 1935 result [1], an instance of the Funk–Hecke theorem for a zonal kernel on S^3; the test below covers $n \le 4$ only. Equation (28) then forces $p_0 = Z/n$, and $E_n = -p_0^2/2 = -Z^2/(2n^2)$ follows with no energy put in. Equivalently, since $1-\Delta_{S^3}$ has eigenvalue n^2 on degree $n-1$, the kernel is $2\pi^2(1-\Delta_{S^3})^{-1/2}$, Eq. (28) reads $p_0\Psi = Z(1-\Delta_{S^3})^{-1/2}\Psi$, and the bound-state Hamiltonian on S^3 is the function $H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$ of the Laplace–Beltrami operator, $E_n = -Z^2/(2(C+1))$ with $C = n^2-1$ the $SO(4)$ Casimir—the precise sense in which the projection makes Δ_{S^3} equivalent

to the Schrödinger problem. **[SYMBOLIC PROOF]** `tests/test_paper7_fock_kernel.py` verifies, in exact arithmetic: the prefactor $Z/(2\pi^2)$ (by a regulated Fourier transform of $-Z/r$); the conformal weights of the reduction to Eq. (28); Eq. (29) at a generic point of S^3 for every (n,l) with $n \le 4$, with one real solid harmonic per l rather than every m (independence of m follows from the kernel's rotation invariance, not from the test) (the harmonics $\sin^l\chi\, C^{l+1}_{n-l-1}(\cos\chi)\, Y_{lm}$ built as harmonic polynomials on $\mathbb{R}^4$, so $l > 0$ is included); the resulting $p_0 = Z/n$ and E_n, with E_n never an input; and that $\Omega^2 \sin^l\chi\, C^{l+1}_{n-l-1}(\cos\chi)$ at $p_0 = 1/n$ is proportional to the independently Fourier-transformed hydrogen radial function for $(n,l) = (1,0), (2,0), (3,0), (2,1)$.

Transcendental content. The π in Eq. (29) is the S^3 measure: at $n = 1$ the eigenvalue is $\mathrm{Vol}(S^3) = 2\pi^2$ (asserted in the same test), read (a reading, not a derivation) as the $k = 0$ volume-slot (M1) π of Paper 18's taxonomy [20]. It cancels exactly against the $1/(2\pi^2)$ of the Fourier-transformed Coulomb kernel, which is why $p_0 = Z/n$ and the levels are rational.

The units (definitional). Given the allowed p_0, the chain to energies is bookkeeping:

1. The S^3 eigenvalue equation: $\Delta_{S^3}\,\Psi_n = -(n^2-1)\,\Psi_n$.
2. The energy-shell constraint: $p_0^2 = -2E$, with $p_0 = Z/n$ from Eq. (29).
3. Combine: for $Z = 1$, $\lambda_n \cdot p_0^2 = -(n^2-1)/n^2 = -(1 - 1/n^2)$.
4. The physical energy: $E_n = -p_0^2/2 = -Z^2/(2n^2)$.

Step 4 is the definition of p_0 rearranged (the appendix's test 18 checks exactly this and is definitional). The Rydberg formula is *not* derived from the sphere's curvature alone: the curvature supplies the integer labels λ_n, the Coulomb kernel's eigenvalue fixes the focal length $p_0 = Z/n$ on each shell, and the energy shell translates that focal length into energy units.

The complete transformation chain is summarized diagrammatically:

$$\boxed{\begin{array}{c} \Delta_{S^3} \text{ on unit } S^3 \\ \text{(labels } (n,l,m)\text{: the graph's node set)} \\ \lambda_n = -(n^2-1) \text{ (dimensionless)} \\ \downarrow \text{ Coulomb kernel eigenvalue } 2\pi^2/n \text{ on shell } n \\ p_0 = Z/n \text{ (focal length, forced)} \\ \downarrow \text{ energy shell } p_0^2 = -2E \\ E_n = -\frac{Z^2}{2n^2} \text{ (dimensionful)} \end{array}} \tag{30}$$

E. Conformal Decomposition of the Wavefunction

We record the form of the S^3 Laplacian acting on a conformally weighted wavefunction.

Define the conformally weighted wavefunction

$$\Phi(p) = \Omega^2(p)\,\varphi(p), \tag{31}$$

where $\varphi(p)$ is the physical momentum-space wavefunction. (This is the weight that the appendix's test 15 uses. It is *not* Fock's S^3 function, which carries the opposite weight, $\Psi = \Omega^{-2}\varphi$, i.e. $\varphi = \Omega^2\Psi$ (Section V D). The second-order statement below holds for either weight, since the coefficient of φ'' is $\Omega^{\pm 2}$ times the nonvanishing metric coefficient.) Applying Δ_{S^3} via the conformal identity Eq. (17), and expanding via the Leibniz rule, one obtains

$$\Delta_{S^3}[\Omega^2\varphi] = \sum_{k=0}^{2} a_k(p)\,\frac{\mathrm{d}^k\varphi}{\mathrm{d}p^k}, \tag{32}$$

where the coefficients $a_k(p)$ are rational functions of p and p_0 (the conformal factor and its derivatives are rational). What the appendix's test 15 verifies is narrower: the coefficient of φ'' is a nonzero rational function, i.e. the operator is second order in the flat momentum coordinate. That check does not discriminate the operator (a wrong connection term also passes it), so it carries none of the equivalence. The equivalence of Δ_{S^3} with the Schrödinger problem ($H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$, Section V D) is carried by Fock's integral equation, Eqs. (27)–(29), whose kernel eigenvalue and eigenfunctions $\varphi_n = \Omega^2\times$ (Fock harmonic) are the tested content (Section V D).

F. Positive Energy: n Released into the Continuum

At positive energy $E = k^2/2$ Fock's projection lands on the unit hyperboloid H^3 instead of S^3, and the compact symmetry $SO(4)$ becomes the non-compact $SO(3,1)$ [12, 13]. **[PRIOR ART]** The release is the continuation $n \to i\eta$, $\eta = Z/k$: the levels $-Z^2/(2n^2)$ continue to $k^2/2$, and the Casimir n^2-1 continues to $-(1+\eta^2)$, minus the spectrum $1+\rho^2$ of the hyperboloid's Laplacian. **[SYMBOLIC PROOF]** The S^3 zonal function $\sin(n\chi)/\sin\chi$, continued by $n \to i\rho$ and $\chi \to i\chi$, is exactly i times the H^3 zonal function $\sin(\rho\chi)/\sinh\chi$, and each satisfies its own eigen-equation. The rotation subgroup $SO(3)$ is untouched, so l, m and their Gaunt selection rules survive. The only thing released is the cap $l \le n-1$: at every $k > 0$ every $l \ge 0$ occurs.

What the continuum adds beyond the continued bound states is one factor. Through Kummer's function the hydrogen radial function is analytic in n, $R_{nl} \propto (2Zr/n)^l e^{-Zr/n} M(l+1-n, 2l+2, 2Zr/n)$, and at $n = i\eta$ it is proportional to the regular Coulomb function. With $dn/dE = n^3/Z^2$, the energy-normalized continuum radial function satisfies

$$R_{El}(r)^2 = \frac{1}{1-e^{-2\pi\eta}}\left[R_{nl}(r)^2\,\frac{dn}{dE}\right]_{n=i\eta} \tag{33}$$

at every r and every l. The factor is the ratio of the two normalizations. It does not depend on l, because the bound normalization's $\prod_{s=-l}^{l}(n+s)$ continues to $i\eta(-1)^l\prod_{s=1}^{l}(s^2+\eta^2)$, and $|\Gamma(l+1+i\eta)|^2 = (\pi\eta/\sinh\pi\eta)\prod_{s=1}^{l}(s^2+\eta^2)$ carries the same product. **[SYMBOLIC PROOF + MEASURED]** The first product identity is checked symbolically and the Gamma identity (classical) at 25 digits, for $l \le 4$; Eq. (33) itself is checked at 20 digits for $l \le 3$, two energies and $Z = 1, 2$.

[PRIOR ART] The factor, and the continuity across threshold of energy-normalized quantities that it expresses, are standard quantum defect theory [11]; what is added here is the reading as one released label.

Two consequences follow. The s-wave contact density per unit energy, $|\psi(0)|^2\,dn/dE$, is exactly Z/π for every bound n, and $(Z/\pi)/(1-e^{-2\pi\eta})$ in the continuum (the Sommerfeld factor [10]). And every oscillator-strength density is the bound closed form continued to $n = i\eta$, times the same factor. For $1s$ the bound side is $f(1s \to np)\,n^3/Z^2 = 2^8 n^8 (n-1)^{2n-4}/(3Z^2(n+1)^{2n+4})$, and its continuation is Stobbe's formula [9, 10],

$$\frac{df}{dE} = \frac{2^8\eta^8\, e^{-4\eta\,\mathrm{arccot}\,\eta}}{3Z^2(1+\eta^2)^4\,(1-e^{-2\pi\eta})}, \tag{34}$$

with both sides tending to $2^8 e^{-4}/(3Z^2)$ at threshold. **[MEASURED]** For $1s \to p$, $2s \to p$, $2p \to s$ and $2p \to d$, at five energies and $Z = 1, 2$, direct quadrature against the regular Coulomb function agrees with the continued bound closed form times the factor to 4×10^{-30}. In every tested transition the radial integral's ${}_2F_1$ terminates after Euler's transformation, leaving $(1-z)^{c-a-b}$ times a polynomial with $1 - z = (n-n_i)/(n+n_i)$; that is the source of the factors $((n-n_i)/(n+n_i))^{2n}$ and, after continuation, of the exponentials $e^{-4\eta\,\mathrm{arccot}(\eta/n_i)}$, so one closed form in n covers both sides of threshold.

Transcendental content. On the bound side every tested oscillator-strength density is rational at each n. The continuum lets in two kinds of number: exponentials of η times arccotangents, whose threshold limits (such as e^{-4}) are already limits of bound-side rationals; and the single factor $1/(1-e^{-2\pi\eta})$, an exponential of π that the bound side never shows and that vanishes faster than any power of E at threshold. No Coulomb phase $\arg\Gamma(l+1+i\eta)$ survives in a squared quantity: at $\eta = 1$ the $1s$ density with the factor and the exponential divided out is exactly $16/3$. Paper 18's ladder [20] has no entry for an exponential of π, so this release adds a number class rather than a rung. The π in Z/π is the solid-angle measure, $Y_{00}^2 = 1/(4\pi)$. The factor has the Bose–Einstein form $1/(1-e^{-x})$ at $x = 2\pi\eta$; no thermal reading of that shared form is made here. The section covers non-relativistic one-electron ions and squared quantities only. Guard: `tests/test_paper7_hydrogen_continuum_door.py`.

Zero energy. Between the sphere and the hyperboloid lies the third shape of Bander and Itzykson's projection, flat space at $E = 0$, reached from the bound side as $n \to \infty$. There the Runge–Lenz algebra contracts: Pauli's commutator $[A_i, A_j] = -2iH\epsilon_{ijk}L_k$ vanishes, $\mathbf{A}^2 = Z^2 + 2H(\mathbf{L}^2+1)$ becomes Z^2, and $SO(4)$ becomes the Euclidean group $E(3)$ [2]. **[SYMBOLIC PROOF + MEASURED]** On the zero-energy states $\sqrt{r}\,J_{2l+1}(\sqrt{8Zr})$ the commutator and $\mathbf{A}^2 - Z^2$ vanish to 10^{-33}, with the operators built symbolically and evaluated at 40 digits; at $E < 0$ the same code keeps $-2iEL_z$. The labels l, m and the Gaunt rules survive, with every l present. Both sides meet one threshold function,

$$\lim_{n\to\infty} R_{nl}(r)^2\,\frac{n^3}{Z^2} = \frac{2\,J_{2l+1}\big(\sqrt{8Zr}\big)^2}{r} = \lim_{k\to0^+} R_{El}(r)^2, \tag{35}$$

approached in whole powers of E from each side, with the same coefficient on both. **[MEASURED]** The bound side is checked for $n \le 1000$ (its $1/n$ coefficient is below 2×10^{-12}) and the continuum for $k \ge 1/800$. This is nearly a corollary of Eq. (33): the bound density per unit energy is a polynomial in $1/n^2$ times the regular solution normalized at the origin, which is entire in E. The threshold function is r^{2l} times a power series in r with rational coefficients, so no π and no exponential of π enter it; the only π at $E = 0$ is the solid angle, and the contact density stays Z/π. **[PRIOR ART]** The analytic approach to $E = 0$ and its Bessel-function limit are also in quantum defect theory [11]. Guard: `tests/test_paper7_zero_energy_door.py`.

VI. TWO-ELECTRON REPULSION ON S^3

We now extend the S^3 framework to the two-electron repulsion integral.

A. Momentum-Space Representation

The direct Coulomb integral (Slater F^0) between orbitals a and b in position space is

$$F^0(a,b) = \int_0^\infty\!\!\int_0^\infty |R_a(r_1)|^2\,|R_b(r_2)|^2\,\frac{1}{r_>}\,r_1^2\,r_2^2\,dr_1\,dr_2\,, \tag{36}$$

where $r_> = \max(r_1, r_2)$. For hydrogen-like orbitals with nuclear charge Z, exact values are known:

$$\begin{aligned} &F^0(1s,1s) = \tfrac{5}{8}Z, \quad F^0(1s,2s) = \tfrac{17}{81}Z, \\ &F^0(2s,2s) = \tfrac{77}{512}Z. \end{aligned} \tag{37}$$

An equivalent momentum-space representation exists for s-orbital pairs. Defining the density Fourier transform

$$\tilde\rho_a(q) = \frac{1}{q}\int_0^\infty |R_a(r)|^2\,r\,\sin(qr)\,dr\,, \tag{38}$$

the Slater integral becomes

$$F^0(a,b) = \frac{2}{\pi}\int_0^\infty \tilde\rho_a(q)\,\tilde\rho_b(q)\,dq\,. \tag{39}$$

For the hydrogen $1s$ and $2s$ orbitals:

$$\tilde{\rho}_{1s}(q) = \frac{16Z^4}{(q^2+4Z^2)^2}\,, \tag{40}$$

$$\tilde{\rho}_{2s}(q) = \frac{Z^4(Z^4-3Z^2q^2+2q^4)}{(Z^2+q^2)^4}\,. \tag{41}$$

B. Fock Projection onto S^3

We apply the Fock stereographic projection with parameter $p_0 = 2Z$, under which $q = 2Z\tan u$ for $u \in [0,\pi/2)$ and $dq = 2Z/\cos^2 u\, du$. Define the *projected density*

$$\Phi_a(t) \equiv \tilde{\rho}_a(2Zt)\,, \tag{42}$$

where $t = \tan u = q/(2Z)$ is the dimensionless momentum transfer. Under this substitution, Eq. (39) becomes

$$\boxed{F^0(a,b) = \frac{4Z}{\pi}\int_0^\infty \Phi_a(t)\,\Phi_b(t)\,dt} \tag{43}$$

[PRIOR ART] This is the master formula: F^0 is a *single* integral over one S^3 coordinate, not a double integral over two points. It is the standard Fourier–Parseval form of the Coulomb integral, $F^0 = (2/\pi)\int_0^\infty \tilde{\rho}_a(q)\tilde{\rho}_b(q)\,dq$, rewritten in the stereographic variable $t = q/(2Z)$; the single-integral structure is Parseval's, not a consequence of the S^3 geometry.

1. $1s$–$1s$ verification

Substituting $q = 2Zt$ into Eq. (40):

$$\Phi_{1s}(t) = \frac{16Z^4}{(4Z^2t^2+4Z^2)^2} = \frac{1}{(t^2+1)^2}\,. \tag{44}$$

Then

$$F^0(1s,1s) = \frac{4Z}{\pi}\int_0^\infty \frac{dt}{(t^2+1)^4} = \frac{4Z}{\pi}\cdot\frac{5\pi}{32} = \frac{5Z}{8}\,. \quad \checkmark \tag{45}$$

In angular coordinates ($t = \tan u$, $dt = du/\cos^2 u$), the integrand becomes $\cos^6 u$: the $1s$ form factor is $\Phi_{1s} = \cos^4 u \propto \Omega^2$, a power of the conformal factor (a renaming in the stereographic variable, not a constant density).

2. $1s$–$2s$ verification (critical test)

Substituting into Eq. (41):

$$\Phi_{2s}(t) = \frac{1-12t^2+32t^4}{(4t^2+1)^4}\,. \tag{46}$$

This is *not* constant on S^3—it has zeros at $t^2 = (3\pm1)/16$ — zeros of the $2s$ density form factor in the momentum transfer, a consequence of the $2s$ radial node rather than a one-to-one image of it (the $2s$ wavefunction has one radial node). The integral evaluates to

$$F^0(1s,2s) = \frac{4Z}{\pi}\int_0^\infty \frac{1-12t^2+32t^4}{(t^2+1)^2(4t^2+1)^4}\,dt = \frac{4Z}{\pi}\cdot\frac{17\pi}{324} = \frac{17Z}{81}\,. \quad \checkmark \tag{47}$$

3. $2s$–$2s$ verification

$$F^0(2s,2s) = \frac{4Z}{\pi}\int_0^\infty \frac{(1-12t^2+32t^4)^2}{(4t^2+1)^8}\,dt = \frac{4Z}{\pi}\cdot\frac{77\pi}{2048} = \frac{77Z}{512}\,. \quad \checkmark \tag{48}$$

[SYMBOLIC PROOF] All three Slater integrals are reproduced exactly by the S^3 density-overlap formula (43) (verified symbolically in `tests/test_paper7_vee_s3.py`: the three integrals evaluate to $5\pi/32$, $17\pi/324$, $77\pi/2048$ in exact arithmetic, reproducing (37)).

C. Structural Result: Node Property vs. Edge Property

The structural reading (of a **[PRIOR ART]** formula: the single-integral structure is Parseval's, and the S^3 variable only renames it) is that Eq. (43) is a *single* integral over one S^3 coordinate (the momentum transfer t), not a double integral over two independent S^3 points. Each orbital a contributes a density *function* $\Phi_a(t)$ on S^3, and V_{ee} is the weighted overlap:

$$V_{ee}(a,b) = \langle\Phi_a|\Phi_b\rangle_{S^3} \equiv \frac{4Z}{\pi}\int_0^\infty \Phi_a(t)\,\Phi_b(t)\,dt\,. \tag{49}$$

In graph-theoretic language, V_{ee} is a **node property** (each node carries a density profile) rather than an **edge property** (a pairwise distance between node coordinates).

a. Failure of the chordal ansatz. A naive chordal-distance ansatz $V_{ee} = \kappa_{ee}/d^2_{\text{chord}}$, where $d^2 = 2(1-\boldsymbol{\xi}_a\cdot\boldsymbol{\xi}_b)$ is the chordal distance between S^3 points assigned to each orbital, gives (with $\kappa_{ee} = 5Z/2$, and the point pairs at $d^2 = 4$, antipodal, for the equal-orbital pairs and $d^2 = 2/5$ for $1s$–$2s$):

Pair	Slater F^0	κ/d^2	Ratio
$1s$–$1s$	$5Z/8$	$5Z/8$	1.00
$1s$–$2s$	$17Z/81$	$25Z/4$	29.8
$2s$–$2s$	$77Z/512$	$5Z/8$	4.2

The chordal ansatz is calibrated to reproduce $F^0(1s,1s)$ exactly (because the $1s$ form factor is a pure power of

the conformal factor, so a single point suffices), but overestimates $F^0(1s, 2s)$ by a factor of 30. The fundamental reason is that the $2s$ density function has nodes and oscillations on S^3 that cannot be captured by a single coordinate point.

D. Natural Projection Scales

Each shell has a natural density form-factor scale, written p_0 here although it is $2Z/n$ — twice the energy-shell focal length Z/n of Section V D (the symbol names two scales):

$$n = 1: \quad p_0 = 2Z \qquad (\text{denominator } q^2 + 4Z^2)\,, \quad (50)$$

$$n = 2: \quad p_0 = Z \qquad (\text{denominator } q^2 + Z^2)\,. \quad (51)$$

When all orbitals are projected using a *single* $p_0 = 2Z$, the $n = 1$ density becomes $\Phi_{1s} = 1/(1 + t^2)^2 \propto \Omega^2$, the square of the conformal factor, while higher-shell densities acquire polynomial modulations. These modulations encode the radial node structure and are responsible for the reduced overlap integrals ($17/81 < 5/8$).

E. Implications

The density-overlap formula (43) has three implications for the computational framework:

1. **Exact Slater integrals in the stereographic variable.** Fourier–Parseval (not the S^3 geometry) converts the position-space double radial integrals into single integrals **[PRIOR ART]**; in the stereographic variable $t = q/(2Z)$ their integrands are rational functions, which can be evaluated in closed form via partial fractions.

2. **Label-lattice implementation.** On the label lattice, each node (orbital label) should carry a density profile $\Phi_a(t)$, and V_{ee} matrix elements should be computed as overlap integrals rather than pairwise edge weights.

3. **Angular momentum extension.** The $1D$ monopole formula (39) applies to s-orbital pairs. For $l > 0$, the full $3D$ angular convolution is required, involving the multipole expansion of the orbital density in spherical harmonics.

VII. GENERALIZATION TO N ELECTRONS

The preceding sections established the S^3 structure for a single electron. The framework generalizes naturally to N-electron atoms: the N-body configuration space $\mathbb{R}^{3N}$ separates, via hyperspherical coordinates, into a hyperradius and the angular unit sphere S^{3N-1} [19] (a polar separation, not a Fock projection).

A. The Angular Manifold and Its Isometry Group

For N electrons at positions $\boldsymbol{r}_1, \ldots, \boldsymbol{r}_N \in \mathbb{R}^3$, the hyperradius $R = \sqrt{\sum_i r_i^2}$ separates a radial coordinate from a $(3N-1)$-dimensional angular manifold. In the limit $R \to 0$ (all electrons at the nucleus), the angular dynamics reduces to the Laplace–Beltrami operator on the unit S^{3N-1}, whose isometry group is SO($3N$).

The eigenvalues of the grand angular momentum operator Λ^2 on S^{3N-1} enter here as the eigenvalues of $\Lambda^2/2$ (the kinetic convention), half the SO($3N$) Casimir eigenvalues $\nu(\nu + 3N - 2)$ of $-\Delta_{S^{3N-1}}$:

$$\mu_{\text{free}}(\nu) = \frac{\nu(\nu + 3N - 2)}{2}, \quad \nu = 0, 1, 2, \ldots \quad (52)$$

These are the "free" eigenvalues—the spectrum before any electron-electron or electron-nuclear interactions are included in the charge function C/R.

B. The Hierarchy of Spheres

Table I summarizes the angular manifold, symmetry group, and Casimir structure for the first few electron numbers. Each row represents a qualitatively different angular problem, but all share the same structural origin: the Laplace–Beltrami operator on a unit sphere of appropriate dimension.

TABLE I. Angular manifold structure for N-electron atoms. The centrifugal term $l_{\text{eff}}(l_{\text{eff}}+1)/(2R^2)$ arises from extracting the Jacobian factor $R^{-(3N-1)/2}$. The effective quantum number $n_{\text{eff}} = l_{\text{eff}} + 1$ determines the quasi-Coulomb energy. The l_{eff} and n_{eff} columns are the free-sphere baseline at $\nu = 0$; for $N \geq 3$ fermionic antisymmetry excludes $\nu = 0$ (the Li ^2S ground state needs $\nu = 2$, Section on fermionic statistics below), so the $N = 3, 4$ rows are not the ground-state channel.

N	S^{d-1}	SO(d)	μ_{free}	l_{eff}	n_{eff}
1	S^2	SO(3)	$\nu(\nu+1)/2$	0	1
2	S^5	SO(6)	$\nu(\nu+4)/2$	3/2	5/2
3	S^8	SO(9)	$\nu(\nu+7)/2$	3	4
4	S^{11}	SO(12)	$\nu(\nu+10)/2$	9/2	11/2

For hydrogen ($N = 1$), S^2 recovers the familiar angular momentum eigenvalues $l(l+1)/2$ (the same halved convention) and the SO(3) rotation group, while S^3 (from the full 3D problem including the radial Fock projection) gives the SO(4) hydrogen symmetry of Section IV. **[PRIOR ART]** For helium ($N = 2$), the angular manifold is S^5 with SO(6) symmetry, and the free eigenvalues $\nu(\nu + 4)/2 = 0, 6, 16, 30, \ldots$ (even-ν singlet sector) are its Casimir values [3]; the hyperspherical angular solver [19] takes them as its $R = 0$ diagonal (`tests/test_algebraic_angular.py` checks the channel bookkeeping: $\nu = 2$, value 6, first appears in the $l = 1$ channel), so they are an input there, not an independent reproduction.

C. Fermionic Statistics and Ground-State Selection

The permutation group S_N acts on the angular manifold by exchanging electron coordinates. Fermionic antisymmetry requires the spatial wavefunction to transform under a specific irreducible representation (irrep) of S_N, determined by the spin state.

For helium ($N = 2$, singlet ^1S), the spatial part is symmetric under S_2 exchange $\alpha \to \pi/2 - \alpha$. This selects even grand angular momentum $\nu = 0, 2, 4, \ldots$, and the ground state sits at $\nu = 0$ with $\mu_{\text{free}} = 0$.

For lithium ($N = 3$, doublet), the spatial part transforms as the standard (mixed-symmetry) irrep $[2,1]$ of S_3. As an S_3 irrep, $[2,1]$ first appears at $\nu = 1$ (the S_N-only Pauli-filling count $\nu_{\min} = 1$ of Paper 16, with $\mu_{\text{free}} = 1 \cdot 8/2 = 4$), but the degree-1 harmonics are the electron coordinates themselves, which carry $L = 1$ and odd parity: they have ^2P, not ^2S, symmetry. The lowest harmonics compatible with $[2,1]$ *and* the ^2S ground state ($L = 0$, even parity) have $\nu = 2$, $\mu_{\text{free}} = 2 \cdot 9/2 = 9$ (the scalar products $\mathbf{r}_i \cdot \mathbf{r}_j$ contain $[2,1]$ twice). Either way the baseline is nonzero, and the higher baseline reflects the Pauli exclusion principle encoded group-theoretically: more electrons require higher angular momentum to satisfy antisymmetry (`tests/test_paper16_nu_min.py::test_Li_Be_L_parity_constrained`).

D. Implications for the Dimensionless Vacuum

The N-electron generalization preserves the core principle: at $R = 0$, the angular dynamics is a free particle on the unit S^{3N-1}, with pure integer-ratio eigenvalues determined by the SO($3N$) Casimir. No physical scale enters. The electron-nuclear and electron-electron interactions enter through the charge function $C(\hat{\Omega})/R$, which acts as a perturbation that lifts the Casimir degeneracy—analogous to how the energy-shell constraint $p_0^2 = -2E$ projects the dimensionless S^3 eigenvalues into dimensionful energies for the single-electron case.

The 18 symbolic checks of Appendix A verify the $N = 1$ case. A generalization to $N > 1$ in which the Laplace–Beltrami operator on S^{3N-1} is the continuum limit of a discrete graph Laplacian on the N-electron angular lattice has no base case, since the $N = 1$ edge Laplacian does *not* converge to Δ_{S^3} (Section III, New Contributions item 1). **[PRIOR ART]** What does generalize is the label-set statement, which is standard hyperspherical-harmonic theory [3]: the angular labels index the harmonics of S^{3N-1}, on which $-\Delta_{S^{3N-1}}$ acts as the diagonal $SO(3N)$ Casimir $\lambda(\lambda + 3N - 2)$.

VIII. DISCUSSION AND CONCLUSIONS

A. Summary of Results

We have established a mathematical chain connecting the discrete GeoVac lattice to the continuous Schrödinger equation through three stages—the first a statement about labels, the second and third exact (Stage 3's kernel eigenvalue carries a factor $2\pi^2$ that cancels against the Coulomb prefactor, leaving rational levels):

Stage 1: Discrete labels ↔ S^3 harmonics. The paraboloid lattice's node set, with n^2-dimensional shells, is exactly the Gel'fand–Tsetlin label set of the Peter–Weyl truncation of $L^2(S^3)$, on which $-\Delta_{S^3} = \mathbf{L}^2 + \mathbf{A}^2$ acts as the diagonal $SO(4)$ Casimir $n^2 - 1$ at every cutoff. **[WALL]** The lattice's edge Laplacian, built from $SU(2) \otimes SU(1,1)$ transition operators, is *not* a discretization of Δ_{S^3}: it preserves l and splits into $n_{\max}$ disconnected blocks (New Contributions item 1).

Stage 2: Curved → Flat. Fock's stereographic projection maps the unit S^3 to flat $\mathbb{R}^3$ momentum space, with conformal factor $\Omega = 2p_0/(p^2 + p_0^2)$. The chordal distance identity (Eq. 14) carries the inverse-square chordal kernel $1/|\boldsymbol{\omega} - \boldsymbol{\omega}'|^2$ on S^3 into the flat-space Coulomb kernel $1/|\mathbf{p} - \mathbf{p}'|^2$ (up to conformal factors). On S^3 that kernel is zonal and diagonal on the harmonics, with eigenvalue $2\pi^2/n$ on degree $n - 1$ (Eq. 29), i.e. it acts as $2\pi^2(1 + \Delta_{S^3})^{-1/2}$; it is not a Green's function of Δ_{S^3} (see Sec. VIII, "Relation to the Coulomb Potential").

Stage 3: Dimensionless → Dimensionful. Fock's S^3 Coulomb kernel has eigenvalue $2\pi^2/n$ on the degree-$(n-1)$ harmonics, which forces the focal length $p_0 = Z/n$; the energy-shell constraint $p_0^2 = -2E$ then converts it into the Rydberg levels $E_n = -Z^2/(2n^2)$ (Section V D). Continued to $n = iZ/k$ the chain reaches the continuum on H^3, whose densities are the continued bound ones times one factor, $1/(1 - e^{-2\pi Z/k})$ (Section V F).

Stage 2 and the bound chain of Stage 3 are verified by symbolic computation; the continuum factor is checked numerically to 20 digits (Section V F). The non-definitional content of Stage 3—the kernel eigenvalue for every (n, l) with $n \leq 4$ (one real harmonic per l; the general-n eigenvalue is Fock's, **[PRIOR ART]**), the resulting $p_0 = Z/n$ with E_n never input, and the identification of $\Omega^2\times$ (Fock harmonic) with the hydrogen momentum wavefunctions—is in `tests/test_paper7_fock_kernel.py`. The 18 appendix tests constitute an algebraic audit of the conformal transformation chain (ten of the eighteen are definitional, limiting-case or non-discriminating checks; the load-bearing identities are items 1, 6, 7, 8, 10, 11, 16 and 17), covering the unit sphere constraint, the conformal factor identities, the projection limits, the chordal distance identity, the dot-product form, the volume Jacobian, the inverse projection, the connection term, the conformal Laplacian of Ω^2, the zero-mode annihilation, the eigenvalue equation for $n = 2$ and $n = 3$, the conformal decomposition, and the (definitional) eigenvalue-to-

energy mapping. Stage 1 is exact as a label statement (the Casimir is built from its generators and checked in `tests/test_graph_s3_nonconvergence.py`); a graph-to-continuum *operator* convergence is refuted, not merely unproven (the saturation of New Contributions item 1(i) has a closed-form rate proved in Paper 0 §VI, but it is a grid band edge; the operator-system rate results of Paper 38 [21] concern a compression of the continuum Camporesi–Higuchi spectral triple, not this graph Laplacian).

B. The Scale-Invariance Property

As Fock showed, conformal projection turns the momentum-space Schrödinger equation for hydrogen into a rotation-invariant integral equation on the unit S^3—a dimensionless, scale-invariant geometry—whose kernel $1/|\boldsymbol{\omega} - \boldsymbol{\omega}'|^2$ is zonal with eigenvalue $2\pi^2/n$ on the degree-$(n-1)$ harmonics (the eigenfunctions of Δ_{S^3}); that eigenvalue forces $p_0 = Z/n$, and since the kernel acts as $2\pi^2(1-\Delta_{S^3})^{-1/2}$ the bound-state Hamiltonian is the function $H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$ of Δ_{S^3} ($E_n = -Z^2/(2(C+1))$, $C = n^2-1$)—the exact sense in which the projection makes Δ_{S^3} equivalent to the Schrödinger problem. What the GeoVac discrete graph shares with this structure is its label set: the nodes (n,l,m) index the unit-S^3 harmonics exactly, and on them $-\Delta_{S^3} = \mathbf{L}^2 + \mathbf{A}^2$ is the diagonal Casimir n^2-1 (the graph's edge Laplacian does not converge to Δ_{S^3}, New Contributions item 1). The continuous equation and its dimensionful energy levels can then be viewed as the projection of those dimensionless labels into observational coordinates. This is a mathematical equivalence, not a proof of physical priority—whether the label set or the differential equation is "more fundamental" is a question of physical interpretation, not of theorem.

Three features of the projection merit emphasis:

(i) Scale invariance. The conformal factor Ω always produces the unit S^3, regardless of p_0. Substituting $u = p/p_0$ yields the standard metric $4/(1+u^2)^2\,\mathrm{d}u^2$ with all p_0-dependence cancelled. The intrinsic geometry of the vacuum has no scale.

(ii) Integer spectrum. The eigenvalues $\lambda_n = -(n^2-1)$ are pure integers determined by the round geometry of the unit S^3; the degeneracy n^2 of each counts the independent oscillation modes at that level, in analogy with standing waves on a finite string.

(iii) Energy as projection parameter. The dimensionful energy $E_n = -Z^2/(2n^2)$ enters exclusively through the energy-shell constraint, with the focal length on each shell, $p_0 = Z/n$, fixed by the Coulomb kernel's eigenvalue. In this reading, energy is the parameter that places the unit sphere in flat coordinates.

C. Relation to the Coulomb Potential

A frequently asked question is: where does the $1/r$ Coulomb potential come from? In the standard formulation, $V(r) = -1/r$ is an input to the Schrödinger equation. As Fock showed [1], in the S^3 formulation it appears instead as the flat-coordinate image of the chordal-distance kernel — an equivalent description, neither one prior.

The mechanism is the chordal distance identity, Eq. (14). On S^3 the relevant kernel is the inverse-square chordal distance $1/|\mathbf{n}-\mathbf{n}'|^2$, the restriction to the sphere of the Green's function of the flat $\mathbb{R}^4$ Laplacian. On S^3 itself it is the zonal kernel with eigenvalue $2\pi^2/n$ on the degree-$(n-1)$ harmonics (Eq. (29)), i.e. proportional to $(1-\Delta_{S^3})^{-1/2}$; it is *not* the Green's function of the Laplace–Beltrami operator, which on a three-manifold is singular like $1/d$ rather than $1/d^2$ and has eigenvalues $\propto 1/(n^2-1)$, divergent at $n=1$. Under stereographic projection, this kernel becomes

$$\frac{1}{|\mathbf{n}-\mathbf{n}'|^2} = \frac{1}{\Omega(\mathbf{p})\,\Omega(\mathbf{p}')\,|\mathbf{p}-\mathbf{p}'|^2}, \tag{53}$$

which, when Fourier transformed to position space, yields the $1/r$ Coulomb potential (with conformal weight corrections that are absorbed into the wavefunction normalization). The Coulomb singularity at $r=0$ corresponds to $|\mathbf{p}| \to \infty$, i.e. to the north pole, the one point where the stereographic map is singular (Eq. (16)); in Fock's description it appears as that coordinate singularity of the projection.

D. Implications for the Discrete Model

The results of this paper bear on the GeoVac framework's computational approach through the label set, not through the edge Laplacian (whose continuum-limit reading is withdrawn, New Contributions item 1). The exact correspondence between the nodes and the S^3 harmonics accounts for the following:

[SYMBOLIC PROOF] *Label exactness:* The algebraic operators reproduce the quantum numbers exactly (Paper 1, a commutator identity): the radial ladder $T_\pm$ generates an $SU(1,1)$ algebra whose Cartan element $N = -2[T_+, T_-]$ returns n on interior shells (Paper 1; the top shell $n_{\max}$ is truncated), and the angular $L_\pm$ generate $SU(2)$ with $L^2 = \ell(\ell+1)$. These are the $SU(1,1)\times SU(2)$ dynamical-symmetry operators, which are distinct from the $SO(4)$ Casimir $\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1$ ($\mathbf{A}$ the energy-normalized Runge–Lenz vector; Pauli [2], Bargmann [4]) that $-\Delta_{S^3}$ is on the labels. The Rydberg energies are not an output of the ladder operators, and the Casimir yields them only through a function that Fock's projection fixes: the eigenvalue $2\pi^2/n$ of Fock's S^3 Coulomb kernel forces $p_0 = Z/n$ and the energy shell $p_0^2 = -2E$ then gives E_n (Stage 3; on the labels $E_n = -Z^2/(2(C+1))$ with

$C = n^2 - 1$, i.e. $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$, the Casimir alone not fixing the function), and the matched scale κ fixes only the graph's band edge, an observation and not a derivation (below).

[**INTERNAL THEOREM**] *Discretization artifacts:* The s/p node-amplitude proxy at finite $n_{\max}$ has a closed-form value on $n_{\max} \equiv 0 \pmod 3$ and decays as $n_{\max} \to \infty$, but its size is set by that residue rather than by boundary effects, and the ℓ-blocks being disconnected, there is no degeneracy for it to lift.

Kinetic scale factor: The constant $\kappa = -1/16$ is the kinetic scale factor that matches the graph's extremal eigenvalue (a band edge) to $-\frac{1}{2}$. Its value 1/16 coincides with a conformal-geometry quantity, as follows. The $l = 0$ eigenfunctions $C^1_{n-1}(\cos\chi)$ of Eq. (25) are the Chebyshev polynomials of the second kind $U_{n-1}(\cos\chi)$, whose three-term recurrence $\cos\chi\, U_{n-1} = \frac{1}{2}U_n + \frac{1}{2}U_{n-2}$ gives a normalized transition amplitude of 1/2 between adjacent shells. The squared inter-shell coupling of $\cos\chi$ in the Gegenbauer eigenbasis is [**PROOF-BY-ARGUMENT + MEASURED**]

$$c^2(n,l) \;:=\; \big|\langle n{+}1, l\,|\cos\chi|\, n, l\rangle\big|^2 \;=\; \frac{1}{4}\left[1 - \frac{l(l+1)}{n(n+1)}\right], \tag{54}$$

(closed form from the Gegenbauer recurrence and norm ratio; verified by quadrature at seven (n,l) pairs, `tests/test_paper7_gegenbauer_coupling.py`). For $l = 0$, $c^2(n,0) = 1/4$ for all n: the amplitude is 1/2, so the coupling is 1/4, and the 1/16 is a *different* quantity. The geometric quantity that the matching κ coincides with is

1. the fourth power of the inverse Fock conformal factor, $1/\Omega^4(0) = 1/16$, where $\Omega(0) = 2$ is the stereographic conformal factor at the projection center ($p = 0$) in the dimensionless $p_0 = 1$ convention (dimensionfully $\Omega(0) = 2/p_0$; the κ observation is stated in the dimensionless convention);

2. equivalently $c^2(n,0)/4$, one quarter of the $l = 0$ base rate of the Casimir decomposition in Eq. (54)—an arithmetic restatement of item 1, not an independent reading.

[**OBSERVATION**] We record this as an *observation*, not a derivation: the energy-matching $\kappa = -1/16$ (the scale mapping the graph's $\lambda_{\max} \to 8$ onto $-\frac{1}{2}$; the 8 is twice the graph's maximum degree, a grid band edge carrying no S^3 information) coincides numerically with the geometric 1/16 above, but the two are computed from disjoint inputs—the energy target and graph coordination on the matching side, the conformal factor on the geometric side—and no mechanism is known that carries one onto the other (a counterfactual energy target moves the matching κ while the geometric 1/16 is rigid; record `tests/test_trunk_qa_kappa.py`, which pins $\lambda_{\max} \to 8$ and $d_{\max} = 4$ and documents the counterfactual, but cannot itself fail on the coincidence). The Z^2 scaling of the hydrogenic levels across nuclear charges is a property of the energy shell ($p_0 = Z/n$), not evidence about the graph's spectrum, which does not carry the Rydberg ladder at any Z.

A notable special value: at $(n,l) = (4,3)$ the Casimir factor is $1 - 12/20 = 2/5$, and $(2/5) \times 1/\Omega^4(0) = 1/40 = \Delta$, the boundary term in Paper 2's Observation-tier combination rule $K = \pi(B + F - \Delta)$ [17]. The value $\Delta = 1/40$ is independently derived from the Dirac degeneracy $g_3^{\text{Dirac}} = 40$ (Paper 2); the composite $[1 - l(l+1)/(n(n+1))]/\Omega^4(0)$ is not a matrix element of anything (the actual coupling is $c^2(4,3) = 1/10$), [**OBSERVATION**] so the coincidence $\Delta = (2/5)/\Omega^4(0)$ is a numerical observation about a composite of two geometric quantities, not a two-route confirmation (`tests/test_trunk_qa_c2_delta.py` records exactly this status). The coupling itself is argued from the Gegenbauer recurrence and verified by quadrature (`tests/test_paper7_gegenbauer_coupling.py`).

E. Limitations

Several caveats apply. First, the single-electron framework of Sections III–V addresses hydrogen only. Section VI extends the framework to two-electron repulsion for s-orbital pairs; the general case with angular momentum requires the full three-dimensional Fourier convolution. Second, the conformal projection presented here recovers only the non-relativistic Schrödinger equation. Relativistic corrections (fine structure, spin-orbit coupling) require a Dirac operator on S^3, the Camporesi–Higuchi operator [23]; its spectral triple is Paper 32 [22]. [**INTERNAL THEOREM**] Third, two rates *are* proved: the saturation deficit $8 - \lambda_{\max} = (C_{\text{sat}} + o(1))/n^2_{\max}$ with $C_{\text{sat}} = 42.7397\ldots$ (Paper 0 §VI), and the s/p proxy's $O(n^{-2}_{\max})$ on $n_{\max} \equiv 0 \pmod 3$, $O(n^{-1}_{\max})$ otherwise (Paper 1 §III). Neither is a rate for an operator convergence $L \to \Delta_{S^3}$, which does not hold (L splits into $n_{\max}$ disconnected l-blocks). The rigorous state-space Gromov–Hausdorff convergence of the scalar truncations for compact metric groups (including $\mathrm{SU}(2) = S^3$) is established by Gaudillot-Estrada and van Suijlekom [24]; for the truncated Camporesi–Higuchi spectral triple on S^3 [23], Paper 38 of this series [21] proves state-space Gromov–Hausdorff convergence under translation-seminorm metrization for this commutative (S^3) factor on the true Camporesi–Higuchi window (convergence of the almost-commutative extension is [**OPEN**] there): the bound $d_{\text{GH}} \le \gamma_n$ and $\gamma_n \to 0$ are proved, with γ_n the Fejér-kernel moment, $(4/\pi + o(1))\log n/n$ in the dual-Coxeter normalization ($2/\pi$ on the unit S^3), [**INTERNAL THEOREM + MEASURED**]; a tapered amplitude gives a [**MEASURED**] $O(1/n)$ (Paper 38, main theorem).

F. Concluding Remarks

Vladimir Fock discovered in 1935 that the hydrogen atom, when viewed in momentum space, lives naturally

on a three-sphere. The scale invariance of this sphere—the fact that it is intrinsically the *unit* S^3 regardless of the energy-shell parameter—is Fock's, and is explicit in Bander and Itzykson's rescaling of momentum onto the unit sphere and hyperboloid [12, 13]; we use it here as an organizing principle. We have articulated this scale-invariance property and **[WALL]** recorded that the GeoVac discrete graph Laplacian does *not* converge to the same S^3 structure: it splits into $n_{\max}$ disconnected l-blocks, and its saturation $\lambda_{\max} \to 2d_{\max}$ is a grid band edge. What the graph shares exactly with the S^3 harmonics is their label set, on which $-\Delta_{S^3} = \mathbf{L}^2 + \mathbf{A}^2$ is the diagonal $SO(4)$ Casimir; the conformal chain from the continuum operator onward is what is audited symbolically (its non-definitional step, the kernel eigenvalue that fixes $p_0 = Z/n$ and makes the Hamiltonian the function $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$, is tested in exact arithmetic for $n \leq 4$; general n is Fock's, **[PRIOR ART]**).

Within this mathematical equivalence, the wave equation can be viewed as a continuous, dimensionful projection of a dimensionless structure whose discrete skeleton is the label set (n, l, m). The Coulomb potential emerges as the coordinate distortion created by the stereographic flattening, and the energy levels emerge from the energy-shell constraint that parameterizes the projection, the Coulomb kernel's eigenvalue $2\pi^2/n$ fixing that parameter to $p_0 = Z/n$ on each shell.

We emphasize that these are statements of mathematical equivalence, not claims about which formulation is physically prior. Whether the topological perspective points toward a deeper theory of quantum mechanics, or provides a computationally useful reformulation, is a question we leave to future investigation. **[SYMBOLIC PROOF]** What is established here is the mathematical chain connecting the continuum S^3 operator—whose harmonics the graph's nodes label—to the continuous equation, audited step by step through symbolic computation (the non-definitional step, the kernel eigenvalue, tested for $n \leq 4$; **[PRIOR ART]** general n, Fock 1935 / Funk–Hecke); the graph's edge Laplacian does not converge to that operator (a wall, New Contributions item 1).

The S^3 conformal equivalence established here is one instance of a broader pattern: separable problems carry analogous label-and-channel constructions, of which the (n, l, m) labels on S^3 are the single-center Coulomb case. A label-set-and-channel construction of the same kind, with different labels, reappears in prolate spheroidal coordinates for one-electron diatomics [18], hyperspherical coordinates for two-electron atoms [19], molecule-frame hyperspherical coordinates for two-electron molecules, and composed fiber bundles for core–valence systems. It is not the same skeleton at every level: at Level 2, for instance, m is carried exactly but the η-node count correlates with ℓ only in the united-atom limit, and the two-center potential lifts the $(2\ell + 1)$-fold degeneracy. The full hierarchy of these embeddings is catalogued in Paper 0 [15].

G. Extension to Molecules

The principle that the natural lattice is the label set and channel structure of the separated coordinate system (not a discretization of its Laplace-Beltrami operator, which the S^3 edge Laplacian is not, New Contributions item 1) extends beyond single-center systems. For diatomic molecules, the two-center Coulomb problem separates in prolate spheroidal coordinates (ξ, η, φ), just as the single-center problem separates in spherical coordinates (leading to S^3 via Fock's projection). Solving the separated prolate spheroidal equations within their channels (labelled by m and the node counts in ξ and η) gives a molecular treatment with R-dependence built into the metric—no fitted parameters are required. **[MEASURED]** For H_2^+, the spectral Laguerre solver of Paper 11 reproduces the two-center reference energy to machine precision (the residual is reference-limited) and gives $R_{\text{eq}} = 1.9973$ bohr on a fine PES grid, with zero free parameters [18] (both values are Paper 11's measurements; the regression test `tests/test_paper11_spectral_radial.py` asserts only the weaker bounds of an error below 0.01% and R_{eq} within 1% of 1.997 bohr; the 0.70% / 2.001 bohr quoted here earlier were Paper 11's superseded finite-difference values). **[CONJECTURE]** This suggests a general principle: every separable quantum system has a natural lattice defined by the coordinate system in which separation occurs.

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Appendix A: Symbolic Test Suite Summary

The following 18 tests were implemented in `sympy` (Python symbolic algebra) and executed with all assertions passing. Each test verifies a specific algebraic identity in the conformal transformation chain. The tests are organized into two modules.

1. Module I: Stereographic Projection Geometry (`tests/test_fock_projection.py`)

1. **Unit sphere constraint.** $\sum_i n_i^2 = 1$ for all $\mathbf{p} \in \mathbb{R}^3$.
2. **Conformal factor identity.** $n_1^2 + n_2^2 + n_3^2 = \Omega^2 |\mathbf{p}|^2$.
3. **South pole limit.** $\mathbf{p} = 0 \Rightarrow \mathbf{n} = (0, 0, 0, -1)$.
4. **North pole limit.** $|\mathbf{p}| \to \infty \Rightarrow \mathbf{n} \to (0, 0, 0, +1)$.
5. **Conformal factor limits.** $\Omega(0) = 2/p_0$; $\Omega(\infty) = 0$.
6. **Chordal distance identity.** $|\mathbf{n} - \mathbf{n}'|^2 = \Omega\,\Omega'\,|\mathbf{p} - \mathbf{p}'|^2$.
7. **Dot product form.** $\mathbf{n} \cdot \mathbf{n}' = 1 - \frac{1}{2}\Omega\,\Omega'\,|\mathbf{p} - \mathbf{p}'|^2$.
8. **Volume Jacobian.** The induced metric of the embedding is $g = J^\top J = \Omega^2\, I_3$ (conformality, all nine entries), hence $\sqrt{\det g} = \Omega^3$; computed from the 4×3 Jacobian of the embedding, not restated.
9. **Numerical spot-check.** All identities verified at concrete floating-point values.
10. **Inverse projection.** $p_i = p_0\, n_i/(1 - n_4)$ round-trips to the identity.

2. Module II: Conformal Laplacian and Eigenvalues (`tests/test_fock_laplacian.py`)

11. **Connection term.** $\mathrm{d}(\ln \Omega)/\mathrm{d}p = -2p/(p^2 + p_0^2)$.
12. **Laplacian of Ω^2.** $\Delta_{\mathrm{flat}}(\Omega^2)/\Omega^2$ is a rational function.
13. **Zero-mode annihilation.** $\Delta_{S^3}(1) = 0$.
14. $n = 2$ **eigenvalue.** $\Delta_{S^3}(\cos\chi) = -3\cos\chi$.
15. **Conformal decomposition.** $\Delta_{S^3}[\Omega^2\varphi]$ has rational coefficients in $\{\varphi, \varphi', \varphi''\}$.
16. $n = 2$ **eigenfunction.** Full eigenvalue equation verified symbolically.
17. $n = 3$ **eigenfunction.** $\Delta_{S^3}(4\cos^2\chi - 1) = -8(4\cos^2\chi - 1)$.
18. **Eigenvalue-to-energy mapping.** $\lambda_n = -(n^2 - 1)$ with $p_0^2 = 1/n^2$ gives $E_n = -1/(2n^2)$ (definitional: the energy-shell relation rearranged, not an independent identity).

3. Supplement: the Coulomb kernel (`tests/test_paper7_fock_kernel.py`)

The non-definitional part of the energy step is tested separately (Section V D): the prefactor $Z/(2\pi^2)$ of Eq. (27); the conformal weights reducing it to Eq. (28); the kernel eigenvalue $2\pi^2/n$, Eq. (29), for all ten (n, l) with $n \leq 4$ at a generic point of S^3, exactly; $\mathrm{Vol}(S^3) = 2\pi^2$ as the $n = 1$ eigenvalue; the resulting $p_0 = Z/n$ and $E_n = -Z^2/(2n^2)$ with no energy input; and $\varphi_n \propto \Omega^2 \sin^l\chi\, C_{n-l-1}^{l+1}(\cos\chi)$ against the Fourier-transformed hydrogen radial functions for $(n, l) = (1, 0), (2, 0), (3, 0), (2, 1)$.

Appendix B: Conformal Laplacian Derivation

For completeness, we derive Eq. (17). Let $\tilde{g}_{ij} = \Omega^2 g_{ij}$ be a conformally rescaled metric in n dimensions, with $\Omega = e^{\omega}$. The Christoffel symbols transform as

$$\tilde{\Gamma}^k_{ij} = \Gamma^k_{ij} + \delta^k_i\,\partial_j\omega + \delta^k_j\,\partial_i\omega - g_{ij}\,g^{kl}\,\partial_l\omega. \tag{B1}$$

The Laplace-Beltrami operator $\Delta_{\tilde{g}} f = \tilde{g}^{ij}\left(\partial_i\partial_j f - \tilde{\Gamma}^k_{ij}\,\partial_k f\right)$ yields, after substitution:

$$\Delta_{\tilde{g}} f = \Omega^{-2}\left[\Delta_g f + (n-2)\,g^{ij}\,(\partial_i\omega)(\partial_j f)\right]. \tag{B2}$$

Setting $n = 3$ and $g = g_{\text{flat}}$ gives Eq. (17), with the connection correction proportional to $(n-2) = 1$.

a. Quadrature-free two-electron FCI. The S^3 density overlap formula (Sec. V) provides exact rational Slater integrals for s-orbital pairs (the general hydrogenic pairs used below are likewise exact rationals, from a separate closed-form route), enabling a fixed-basis FCI calculation of a two-electron model with no numerical quadrature. Using a *hybrid* one-body operator — the exact hydrogenic eigenvalues $-Z^2/2n^2$ on the diagonal and the graph Laplacian's off-diagonal entries scaled by κ, the degree term dropped (`geovac/casimir_ci.py`) — together with rational Slater integrals as the two-body operator, a two-electron ground state is computed with *no fitted parameters*: no orbital exponent, no variational optimization, no numerical quadrature. The exact one-electron spectrum is an *input* here rather than a graph output — $\kappa\mathcal{L}$ alone has $\lambda_{\max} = 0$ at $n_{\max} = 1$ and would give $E = +5Z/8$, not the value below — so this calculation is not a demonstration that the graph supplies the one-body physics. The FCI matrix at each $n_{\max}$ is a rational matrix whose ground-state eigenvalue is an algebraic number.

The $n_{\max} = 7$ (1,218 singlet configurations) energy of this hybrid operator is not a helium result: the off-diagonal graph entries are not part of helium's Hamiltonian; in the $k = Z$ hydrogenic basis the one-body operator $T + V_{ne}$ is exactly diagonal, and the off-diagonal κ-scaled entries are an added hop of $+\frac{1}{16}$ per edge that lowers the energy by 48.3 mHa at $n_{\max} = 7$ (49.2 mHa at $n_{\max} = 4$)—more than helium's whole correlation energy (42.0 mHa). That energy is the ground state of the hop-augmented model, not a variational bound on helium. With helium's physical Hamiltonian in the same basis (diagonal one-body operator, the same rational Slater integrals) the **[MEASURED]** error at $n_{\max} = 7$ is 1.88% above exact (the ladder $n_{\max} = 2$–6 runs 2.422, 2.094, 1.981, 1.928, 1.899%; the $n_{\max} = 5$ point and its 48.74 mHa hop shift are pinned in `tests/test_he_physical_hamiltonian.py::test_paper7_he_nmax` the $n_{\max} = 2$–4 points in the same file; the $n_{\max} = 6$ and 7 points and the 48.3/49.2 mHa hop shifts are measured values not re-run in the suite), a floor set by the missing continuum of the fixed-k bound-state basis (it lies above the Hartree–Fock limit), not by the cusp; see Paper 13, Sec. "Conclusion."

FCI basis invariance is checked computationally: transforming V_{ee} to the graph eigenbasis reproduces the ground-state energy to 10^{-10} Ha at $n_{\max} = 1$–3 and the whole spectrum to 10^{-8} Ha at $n_{\max} = 3$ (`tests/test_casimir_ci.py`, `TestGraphConsistentCI`). This says nothing about which one-body operator is physical, nor about the origin of the convergence floor.

At $n_{\max} = 1$ (a single node, so no edges and no hop), the fixed-k CI energy reduces to **[SYMBOLIC PROOF]** $E = -Z^2 + 5Z/8 = -11/4$ Ha for $Z = 2$, an exact rational (two one-body terms $-Z^2/2$ plus the repulsion $F^0(1s, 1s) = 5Z/8$). **[SYMBOLIC PROOF]** The variationally optimized single-exponent result $k^* = 27/16$, $E^* = -729/256$ Ha is also an exact rational. A 2D variational solver treating hyperradius and hyperangle simultaneously (Paper 13, Note added v2.6.0) reaches a properly variational 0.022% at $l_{\max} = 7$; a Schwartz cusp extrapolation at $l_{\max} = 4$ gives 0.020%, still above the exact energy (non-variational, Paper 13).